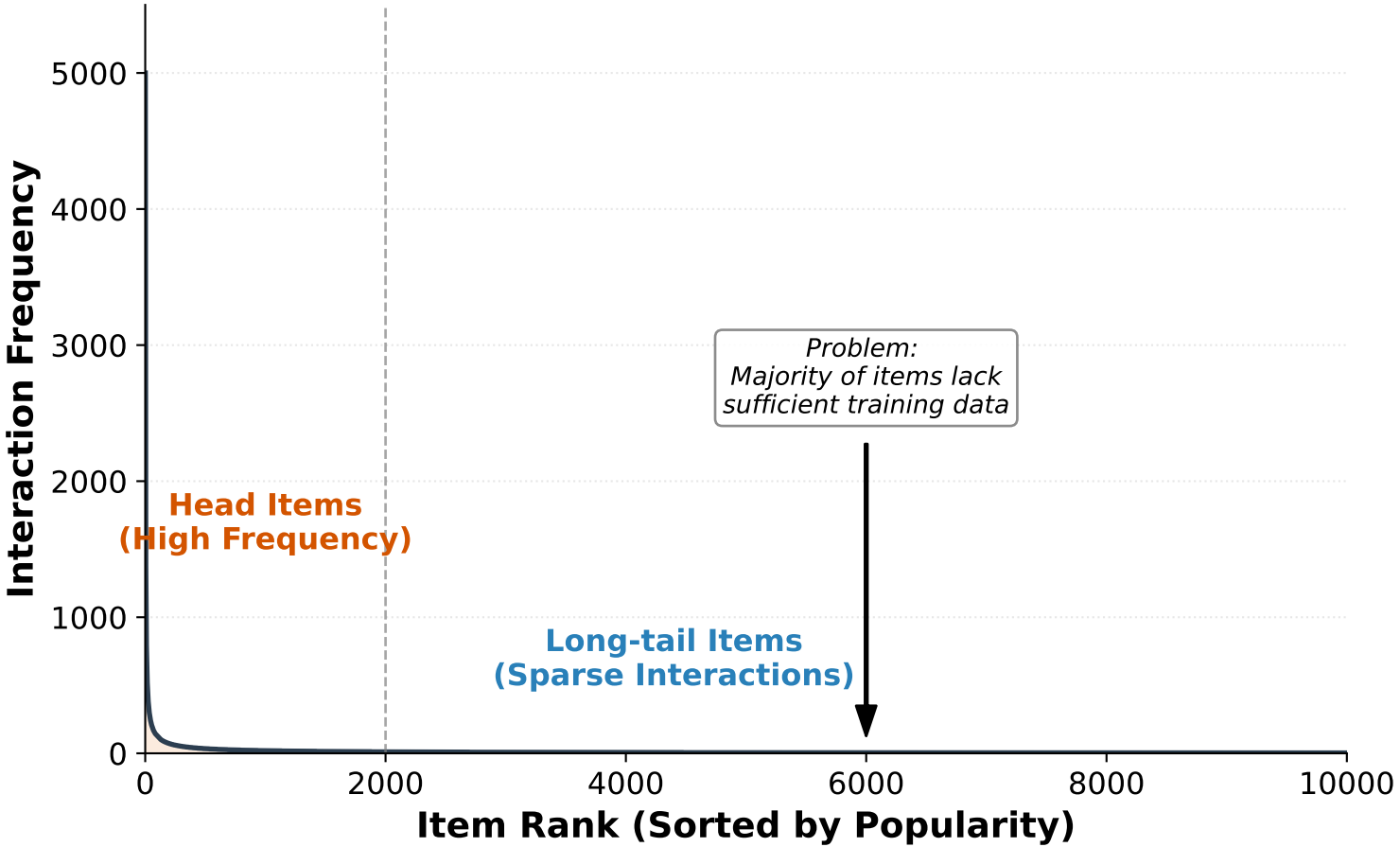

Interaction Frequency
5000
4000
3000
2000
1000
0
0
2000
4000
6000
8000
10000
Item Rank (Sorted by Popularity)
Head Items
(High Frequency)
Long-tail Items
(Sparse Interactions)
Problem:
Majority of items lack
sufficient training data